

```
-- create database
CREATE DATABASE data_transformer;
-- use database
USE data_transformer;

-- Create customers Table
CREATE TABLE CUSTOMERS (
    CustomerID INT PRIMARY KEY,
    FirstName VARCHAR(25) NOT NULL,
    LastName VARCHAR(25),
    Email VARCHAR(50) UNIQUE,
    RegistrationDate DATE
);

-- Create orders Table
CREATE TABLE ORDERS (
    OrderID INT PRIMARY KEY,
    CustomerID INT,
    OrderDate DATE,
    TotalAmount DECIMAL(10,2),
    FOREIGN KEY (CustomerID) REFERENCES CUSTOMERS(CustomerID)
);

-- Create employees Table
CREATE TABLE EMPLOYEES (
    EmployeeID INT PRIMARY KEY,
    FirstName VARCHAR(25) NOT NULL,
    LastName VARCHAR(25),
    Department VARCHAR(50),
    HireDate DATE,
    Salary DECIMAL(10,2)
);

-- Insert customers records
INSERT INTO CUSTOMERS (CustomerID, FirstName, LastName, Email, RegistrationDate) VALUES
(1, 'John', 'Doe', 'john.doe@email.com', '2022-03-15'),
(2, 'Jane', 'Smith', 'jane.smith@email.com', '2021-11-02'),
(3, 'Ritu', 'Shah', 'ritu.shah@email.com', '2020-05-03'),
(4, 'Pratik', 'Khan', 'pratik.khan@email.com', '2021-12-05'),
(5, 'Ekta', 'Mehta', 'ekta.mehta@email.com', '2022-08-06'),
(6, 'Moni', 'Patel', 'moni.patel@email.com', '2023-11-07'),
(7, 'Rinku', 'Shah', 'rinku.shah@email.com', '2021-03-18'),
(8, 'Akash', 'Jain', 'akash.jain@email.com', '2023-05-29'),
(9, 'Aditi', 'Parmar', 'aditi.parmar@email.com', '2022-06-14'),
(10, 'Dev', 'Vyas', 'dev.vyas@email.com', '2022-09-13'),
(11, 'Rajat', 'Koli', 'rajat.koli@email.com', '2020-01-16'),
(12, 'Kartik', 'Mehra', 'kartik.mehra@email.com', '2022-09-13');
```

```
-- Insert Orders records
INSERT INTO ORDERS (OrderID, CustomerID, OrderDate, TotalAmount) VALUES
(101, 1, '2023-07-01', 150.50),
(102, 2, '2023-07-03', 200.75),
(103, 8, '2023-03-05', 1200.00),
(104, 4, '2023-04-04', 750.00),
(105, 5, '2023-05-25', 300.00),
(106, 6, '2023-06-12', 950.00),
(107, 7, '2023-07-18', 1100.00),
(108, 3, '2023-08-16', 450.00),
(109, 10, '2023-09-26', 600.00),
(110, 9, '2023-10-23', 1300.00),
(111, 12, '2023-11-12', 2000.00),
(112, 11, '2023-12-19', 520.00),
(113, 3, '2023-08-25', 2000.00),
(114, 4, '2023-05-24', 520.00);
```

```
-- Insert Employees records
INSERT INTO EMPLOYEES (EmployeeID, FirstName, LastName, Department, HireDate, Salary) VALUES
(1, 'Mark', 'Johnson', 'Sales', '2020-01-15', 50000.00),
(2, 'Susan', 'Lee', 'HR', '2021-03-20', 55000.00),
(3, 'Rehan', 'Gupta', 'Sales', '2023-04-05', 60000.00),
(4, 'Pooja', 'Chaudhari', 'IT', '2021-06-21', 75000.00),
(5, 'Veena', 'Parmar', 'Finance', '2023-05-25', 56000.00),
(6, 'Shreya', 'Patel', 'IT', '2020-12-30', 78000.00),
(7, 'Ayushi', 'Yadav', 'Sales', '2022-07-16', 81000.00),
(8, 'Khushi', 'Parmar', 'HR', '2019-04-25', 55000.00),
(9, 'Devvanshi', 'Joshi', 'Finance', '2020-03-29', 35000.00),
(10, 'Riya', 'Patel', 'Sales', '2023-01-13', 80000.00),
(11, 'Alvish', 'Yadav', 'HR', '2019-10-08', 25000.00),
(12, 'Krihsa', 'Pal', 'Finance', '2022-08-05', 75000.00);
```

```
SELECT * FROM CUSTOMERS;
-- SELECT COUNT(*) FROM CUSTOMERS;
```

CustomerID	FirstName	LastName	Email	RegistrationDate
1	John	Doe	john.doe@email.com	2022-03-15
2	Jane	Smith	jane.smith@email.com	2021-11-02
3	Ritu	Shah	ritu.shah@email.com	2020-05-03
4	Pratik	Khan	pratik.khan@email.com	2021-12-05
5	Ekta	Mehta	ekta.mehta@email.com	2022-08-06
6	Moni	Patel	moni.patel@email.com	2023-11-07
7	Rinku	Shah	rinku.shah@email.com	2021-03-18
8	Akash	Jain	akash.jain@email.com	2023-05-29
9	Aditi	Parmar	aditi.parmar@email.com	2022-06-14
10	Dev	Vyas	dev.vyas@email.com	2022-09-13
TOTAL	TOTAL	TOTAL	TOTAL	TOTAL

```
SELECT * FROM ORDERS;
```

OrderID	CustomerID	OrderDate	TotalAmount
101	1	2023-07-01	150.50
102	2	2023-07-03	200.75
103	8	2023-03-05	1200.00
104	4	2023-04-04	750.00
105	5	2023-05-25	300.00
106	6	2023-06-12	950.00
107	7	2023-07-18	1100.00
108	3	2023-08-16	450.00
109	10	2023-09-26	600.00
110	9	2023-10-23	1300.00
TOTAL	TOTAL	TOTAL	TOTAL

```
SELECT * FROM EMPLOYEES;
```

EmployeeID	FirstName	LastName	Department	HireDate	Salary
1	Mark	Johnson	Sales	2020-01-15	50000.00
2	Susan	Lee	HR	2021-03-20	55000.00
3	Rehan	Gupta	Sales	2023-04-05	60000.00
4	Pooja	Chaudhari	IT	2021-06-21	75000.00
5	Veena	Parmar	Finance	2023-05-25	56000.00
6	Shreya	Patel	IT	2020-12-30	78000.00
7	Ayushi	Yadav	Sales	2022-07-16	81000.00
8	Khushi	Parmar	HR	2019-04-25	55000.00
9	Devvanshi	Joshi	Finance	2020-03-29	35000.00
10	Riya	Patel	Sales	2023-01-13	80000.00
NULL	NULL	NULL	NULL	NULL	NULL

```
-- 1. INNER JOIN: Retrieve all orders and customer details where orders exist.
```

```
SELECT O.OrderId,C.FirstName,C.LastName, O.OrderDate, O.TotalAmount
FROM ORDERS AS O
INNER JOIN CUSTOMERS AS C
ON O.CustomerID = C.CustomerID;
```

EmployeeID	FirstName	LastName	Department	HireDate	Salary
1	Mark	Johnson	Sales	2020-01-15	50000.00
2	Susan	Lee	HR	2021-03-20	55000.00
3	Rehan	Gupta	Sales	2023-04-05	60000.00
4	Pooja	Chaudhari	IT	2021-06-21	75000.00
5	Veena	Parmar	Finance	2023-05-25	56000.00
6	Shreya	Patel	IT	2020-12-30	78000.00
7	Ayushi	Yadav	Sales	2022-07-16	81000.00
8	Khushi	Parmar	HR	2019-04-25	55000.00
9	Devvanshi	Joshi	Finance	2020-03-29	35000.00
10	Riya	Patel	Sales	2023-01-13	80000.00
NULL	NULL	NULL	NULL	NULL	NULL

```
-- 2. LEFT JOIN: Retrieve all customers and their corresponding orders (if any).
```

```
SELECT O.OrderId,C.FirstName,C.LastName, O.OrderDate, O.TotalAmount
FROM CUSTOMERS AS C
LEFT JOIN ORDERS AS O
ON C.CustomerID = O.CustomerID;
```

OrderId	FirstName	LastName	OrderDate	TotalAmount
101	John	Doe	2023-07-01	150.50
102	Jane	Smith	2023-07-03	200.75
108	Ritu	Shah	2023-08-16	450.00
113	Ritu	Shah	2023-08-25	2000.00
104	Pratik	Khan	2023-04-04	750.00
114	Pratik	Khan	2023-05-24	520.00
105	Ekta	Mehta	2023-05-25	300.00
106	Moni	Patel	2023-06-12	950.00
107	Rinku	Shah	2023-07-18	1100.00
103	Akash	Jain	2023-03-05	1200.00

```
-- 3. RIGHT JOIN: Retrieve all orders and their corresponding customers (if any).
```

```
SELECT O.OrderId,C.FirstName,C.LastName, O.OrderDate, O.TotalAmount
FROM CUSTOMERS AS C
RIGHT JOIN ORDERS AS O
ON O.CustomerID = C.CustomerID;
```

OrderId	FirstName	LastName	OrderDate	TotalAmount
101	John	Doe	2023-07-01	150.50
102	Jane	Smith	2023-07-03	200.75
103	Akash	Jain	2023-03-05	1200.00
104	Pratik	Khan	2023-04-04	750.00
105	Ekta	Mehta	2023-05-25	300.00
106	Moni	Patel	2023-06-12	950.00
107	Rinku	Shah	2023-07-18	1100.00
108	Ritu	Shah	2023-08-16	450.00
109	Dev	Vyas	2023-09-26	600.00
110	Aditi	Parmar	2023-10-23	1300.00

```
-- 4. FULL OUTER JOIN: Retrieve all customers and all orders, regardless of matching.
SELECT C.FirstName, C.LastName, O.OrderDate, O.TotalAmount
FROM CUSTOMERS AS C
LEFT JOIN ORDERS AS O
ON C.CustomerID = O.CustomerID
UNION
SELECT C.FirstName, C.LastName, O.OrderDate, O.TotalAmount
FROM CUSTOMERS AS C
RIGHT JOIN ORDERS AS O
ON C.CustomerID = O.CustomerID;
```

FirstName	LastName	OrderDate	TotalAmount
John	Doe	2023-07-01	150.50
Jane	Smith	2023-07-03	200.75
Ritu	Shah	2023-08-16	450.00
Ritu	Shah	2023-08-25	2000.00
Pratik	Khan	2023-04-04	750.00
Pratik	Khan	2023-05-24	520.00
Ekta	Mehta	2023-05-25	300.00
Moni	Patel	2023-06-12	950.00
Rinku	Shah	2023-07-18	1100.00
Akash	Jain	2023-03-05	1200.00
Aditi	Parmar	2023-10-23	1300.00
Dev	Vyas	2023-09-26	600.00
Rajat	Koli	2023-12-19	520.00
Kartik	Mehara	2023-11-12	2000.00

```
-- 5. Subquery to find customers who have placed orders worth more than the average amount.
SELECT DISTINCT C.CustomerID, C.FirstName, C.LastName
FROM CUSTOMERS C
JOIN ORDERS O
ON C.CustomerID = O.CustomerID
WHERE O.TotalAmount > (SELECT AVG(TotalAmount) FROM ORDERS);
```

CustomerID	FirstName	LastName
8	Akash	Jain
6	Moni	Patel
7	Rinku	Shah
9	Aditi	Parmar
12	Kartik	Mehara
3	Ritu	Shah

```
-- 6. Subquery to find employees with salaries above the average salary.
SELECT EmployeeId, FirstName, Salary FROM EMPLOYEES
WHERE Salary > (SELECT AVG(Salary) FROM EMPLOYEES);
```

EmployeeId	FirstName	Salary
4	Pooja	75000.00
6	Shreya	78000.00
7	Ayushi	81000.00
10	Riya	80000.00
12	Krihsa	75000.00

```
-- 7. Extract the year and month from the OrderDate.
SELECT OrderID, YEAR(OrderDate) AS Order_Year, MONTH(OrderDate) AS Order_Month
FROM ORDERS;
```

OrderID	Order_Year	Order_Month
101	2023	7
102	2023	7
103	2023	3
104	2023	4
105	2023	5
106	2023	6
107	2023	7
108	2023	8
109	2023	9
110	2023	10

```
-- 8. Calculate the difference in days between two dates (order date and current date).
SELECT OrderID, DATEDIFF(CURDATE(), OrderDate) AS Days_Difference
FROM ORDERS;
```

OrderID	Days_Difference
101	967
102	965
103	1085
104	1055
105	1004
106	986
107	950
108	921
109	880
110	853

```
-- 9. Format the OrderDate to a more readable format (e.g., 'DD-MMM-YYYY').
SELECT OrderID, DATE_FORMAT(OrderDate, '%d-%b-%Y') AS Formatted_Date
FROM ORDERS;
```

OrderID	Formatted_Date
101	01-Jul-2023
102	03-Jul-2023
103	05-Mar-2023
104	04-Apr-2023
105	25-May-2023
106	12-Jun-2023
107	18-Jul-2023
108	16-Aug-2023
109	26-Sep-2023
110	23-Oct-2023

```
-- 10. Concatenate FirstName and LastName to form a full name.
SELECT CustomerId, CONCAT(FirstName, ' ', LastName) AS FullName
FROM CUSTOMERS;
```

CustomerId	FullName
1	John Doe
2	Jane Smith
3	Ritu Shah
4	Pratik Khan
5	Ekta Mehta
6	Moni Patel
7	Rinku Shah
8	Akash Jain
9	Aditi Parmar
10	Dev Vyas

```
-- 11. Replace part of a string (e.g., replace 'John' with 'Jonathan').
SELECT FirstName, REPLACE(FirstName, 'John', 'Jonathan') AS UpdatedName
FROM CUSTOMERS;
```

FirstName	UpdatedName
John	Jonathan
Jane	Jane
Ritu	Ritu
Pratik	Pratik
Ekta	Ekta
Moni	Moni
Rinku	Rinku
Akash	Akash
Aditi	Aditi
Dev	Dev

```
-- 12. Convert FirstName to uppercase and LastName to lowercase.
SELECT CustomerId, UPPER(FirstName) AS UpperFName, LOWER(LastName) AS LowerLName
FROM CUSTOMERS;
```

CustomerId	UpperFName	LowerLName
1	JOHN	doe
2	JANE	smith
3	RITU	shah
4	PRATIK	khan
5	EKTA	mehta
6	MONI	patel
7	RINKU	shah
8	AKASH	jain
9	ADITI	parmar
10	DEV	vyas

```
-- 13. Trim extra spaces from the Email field.
SELECT TRIM(Email) AS CleanEmail
FROM CUSTOMERS;
```

CleanEmail
ritu.shah@email.com
aditi.parmar@email.com
akash.jain@email.com
dev.vyas@email.com
ekta.mehta@email.com
jane.smith@email.com
john.doe@email.com
kartik.mehra@email.com
moni.patel@email.com
pratik.khan@email.com

```
-- 14. Calculate the running total of TotalAmount for each order.
SELECT OrderID, OrderDate, TotalAmount, SUM(TotalAmount)
OVER(ORDER BY OrderDate) AS RunningTotal
FROM ORDERS;
```

OrderID	OrderDate	TotalAmount	RunningTotal
103	2023-03-05	1200.00	1200.00
104	2023-04-04	750.00	1950.00
114	2023-05-24	520.00	2470.00
105	2023-05-25	300.00	2770.00
106	2023-06-12	950.00	3720.00
101	2023-07-01	150.50	3870.50
102	2023-07-03	200.75	4071.25
107	2023-07-18	1100.00	5171.25
108	2023-08-16	450.00	5621.25
113	2023-08-25	2000.00	7621.25
109	2023-09-26	600.00	8221.25
110	2023-10-23	1300.00	9521.25
111	2023-11-12	2000.00	11521.25
112	2023-12-19	520.00	12041.25

```
-- 15. Rank orders based on TotalAmount using the RANK() function.
SELECT OrderID, TotalAmount,
RANK() OVER(ORDER BY TotalAmount DESC) AS RankOrder
FROM ORDERS;
```

OrderID	TotalAmount	RankOrder
111	2000.00	1
113	2000.00	1
110	1300.00	3
103	1200.00	4
107	1100.00	5
106	950.00	6
104	750.00	7
109	600.00	8
112	520.00	9
114	520.00	9
108	450.00	11
105	300.00	12
102	200.75	13
101	150.50	14

```
-- 16. Assign a discount based on TotalAmount in orders (e.g., > 1000: 10% off, > 500: 5% off).
SELECT OrderID, TotalAmount,
CASE
WHEN TotalAmount > 1000 THEN '10% off'
WHEN TotalAmount > 500 THEN '5% off'
ELSE 'No Discount'
END AS Discount
FROM ORDERS;
```

OrderID	TotalAmount	Discount
101	150.50	No Discount
102	200.75	No Discount
103	1200.00	10% off
104	750.00	5% off
105	300.00	No Discount
106	950.00	5% off
107	1100.00	10% off
108	450.00	No Discount
109	600.00	5% off
110	1300.00	10% off

```
-- 17. Categorize employees' salaries as high, medium, or low.
SELECT FirstName, Salary,
CASE
WHEN Salary >= 75000 THEN 'High'
WHEN Salary >= 45000 THEN 'Medium'
ELSE 'Low'
END AS Salary_Category
FROM EMPLOYEES;
```

FirstName	Salary	Salary_Category
Mark	50000.00	Medium
Susan	55000.00	Medium
Rehan	60000.00	Medium
Pooja	75000.00	High
Veena	56000.00	Medium
Shreya	78000.00	High
Ayushi	81000.00	High
Khushi	55000.00	Medium
Devyanshi	35000.00	Low
Riya	80000.00	High